

English Verbs as a Semantic Net

Christiane Fellbaum

This paper describes the semantic network of English verbs in WordNet. The semantic relations used to build networks of nouns and adjectives cannot be applied without modification, but have to be adapted to fit the semantics of verbs, which differ substantially from those of the other lexical categories. The nature of these relations is discussed, as is their distribution throughout different semantic groups of verbs, which determines certain idiosyncratic patterns of lexicalization. In addition, four variants of lexical entailment are distinguished, which interact in systematic ways with the semantic relations. Finally, the lexical properties of the different verb groups are outlined.

Verbs are arguably the most important lexical and syntactic category of a language. All English sentences must contain at least one verb, but, as grammatical sentences with “dummy” subjects like *It is snowing* show, they need not contain a (referential) noun. Many linguists have argued for a model of sentence meaning in which verbs occupy the core position and function as the central organizers of sentences (Chafe, 1970; Fillmore, 1968; and others). The verb provides the relational and semantic framework for its sentence. Its predicate-argument structure (or subcategorization frame) specifies the possible syntactic structures of the sentences in which it can occur. The linking of noun arguments with thematic roles or cases, such as INSTRUMENT, determines the different meanings of the events or states denoted by the sentence, and the selectional restrictions specify the semantic properties of the noun classes that can flesh out the frame. This syntactic and semantic information is generally thought to be part of the verb’s lexical entry, that is to say, part of the information about the verb that is stored in a speaker’s mental lexicon. Because of the complexity of this information, verbs are probably the lexical category that is most difficult to study.

Polysemy

Even though grammatical English sentences require a verb though not necessarily a noun, the language has far fewer verbs than nouns. For example, the *Collins English Dictionary* lists 43,636 different nouns and 14,190 different verbs. Verbs are more polysemous than nouns: the nouns in *Collins* have on the average 1.74 senses, whereas verbs average 2.11 senses.²

The higher polysemy of verbs suggests that verb meanings are more flexible than noun meanings. Verbs can change their meanings depending on the kinds of noun arguments with which they co-occur, whereas the meanings of nouns tend to be more stable in the presence of different verbs. Gentner and France (1988) have demonstrated what they call the high mutability of verbs. They presented subjects with sentences

² We are indebted to Richard Beckwith for computing these figures.

containing verbs in conjunction with nouns that violated the verbs' selectional restrictions. When asked to paraphrase the sentences, subjects assigned novel interpretations to the verbs, but did not modify the literal meanings of the nouns. Gentner and France concluded that verb meanings are more easily altered because they are less cohesive than those of nouns—a flexibility that makes a semantic analysis of verbs an even more challenging task.

The most frequently used verbs (*have, be, run, make, set, go, take*, and others) are also the most polysemous, and their meanings often depend heavily on the nouns with which they co-occur. For example, dictionaries differentiate between the senses of *have* in sentences like *I have a Mercedes* and *I have a headache*. The difference is less due to the polysemy of *have*, however, than to the concrete or abstract nature of its objects.

In the case of such polysemous verbs as *beat*, meaning differences are determined less by the semantics of the verb's arguments than by different elaborations of one or two common core components shared by most senses of *beat*. Different senses of *beat* occur in very different semantic domains: {*beat, strike, hit*} is a contact verb; {*beat, flatten*} is a verb of change; {*beat, throb, pulse*} is a motion verb; {*beat, defeat*} is a competition verb; {*beat, flog, punish*} and {*beat, circumvent (the system)*} are verbs in the domain of social interaction; {*beat, shape, do metalwork*} is a creation verb; {*beat, baffle*} is a cognition verb; {*beat, stir, whisk*} belongs to the domain of cooking verbs; and {*beat, mark*} is a kind of motion performed to indicate the counts in music. Although most of these verbs seem to share a semantic component of CONTACT or IMPACT, the differences illustrate how flexible these core meanings can be.

In order to reduce ambiguity in WordNet, verb synsets could contain cross-reference pointers to the noun synsets that contain nouns selected for by the verbs. For example, one sense of the verb *throw* (*throw on a wheel*) always selects the noun *pottery* or its hyponyms as its object; that selectional restriction could be represented by a labeled pointer. At the present time, however, this possibility has not been implemented in WordNet.

The Organization of Verbs in WordNet

Currently, WordNet contains over 21,000 verb word forms (of which over 13,000 are unique strings) and approximately 8,400 word meanings (synsets). Included are phrasal verbs like *look up* and *fall back*.

Verbs are divided into 15 files, largely on the basis of semantic criteria. All but one of these files correspond to what linguists have called semantic domains: verbs of bodily care and functions, change, cognition, communication, competition, consumption, contact, creation, emotion, motion, perception, possession, social interaction, and weather verbs. Virtually all the verbs in these files denote events or actions. Another file contains verbs referring to states, such as *suffice, belong*, and *resemble*, that could not be integrated into the other files. The verbs in this latter group do not constitute a semantic domain, and share no semantic properties other than that they refer to states. This file, whose organization resembles that of the adjectives in WordNet, consists of small semantic clusters. The division of verbs into 14 files corresponding to different semantic

domains, each containing event and action verbs, and one file containing semantically diverse stative verbs reflects the division between the major conceptual categories EVENT and STATE found in Jackendoff's (1983:170) and Dowty's (1979) analyses. The boundaries between the files are not rigid, and the particular classification was chosen largely because it permits us to get a grasp on the organization of the verbs; it has no further theoretical or psychological implications.

Many of the files derive their names from the topmost verbs, or "unique beginners," which head these semantically coherent lexical groups. These topmost verbs resemble the "core components" of Miller and Johnson-Laird (1976). They are the unelaborated concepts from which the verbs constituting the semantic field are derived via semantic relations.

Synonymy

Few truly synonymous verbs, such as *shut* and *close*, can be found in the lexicon—the number depends on how loose a definition of synonymy one adopts. The best examples are probably verb concepts that are represented by both an Anglo-Saxon and a Greco-Latinate word: *begin-commence*, *end-terminate*, *rise-ascend*, *blink-nictate*, *behead-decapitate*, *spit-expectorate*. In general, the Greco-Latinate verbs are used in more formal or technical speech registers: *buy* vs. *purchase*, *sweat* vs. *perspire*, or *shave* vs. *epilate*. Cruse (1986:268) points out that frequently only one member of such a synonym pair tends to be felicitous in a given context: *Where have you hidden Dad's slippers?* sounds more natural than *Where have you concealed Dad's slippers?* And subtle meaning differences can show up in different selectional restrictions. For example, *rise* and *fall* can select as an argument such abstract entities as *the temperature*, but their close synonyms *ascend* or *descend* cannot. Because many apparently synonymous verbs exhibit such differences, verb synsets in WordNet often contain periphrastic expressions, rather than lexicalized synonyms.

These periphrases break down a synonymous verb into an entire verb phrase and thereby often reflect the way in which the verb has become lexicalized by showing constituents that have been conflated in the verb. For example, a denominal verb such as *hammer* is listed with the parenthetical gloss *hit with a hammer*; the conflated verb denotes the function of the noun. The periphrases indicate the basic action and the role of the noun (material and instrument) with which the action is performed. The synonymous expressions of deadjectival verbs often have the form *make* or *become* + some adjective: {*whiten*, *become white*}, {*enrich*, *make rich*}, etc. Thus they reflect the fact that these verbs are for the most part verbs of change. The synonymous expressions of many verbs show that they are manner elaborations of a more basic verb: {*swim*, *travel through water*}, {*mumble*, *talk indistinctly*}, {*saute*, *fry briefly*}, etc.

Representing the meanings of verbs is difficult for any theory of lexical semantics, but especially so for WordNet, which differs from previous approaches in avoiding a semantic decomposition in favor of a relational analysis.

Decompositional vs. Relational Semantic Analysis

Most approaches to verb semantics have been attempts at decomposition in one form or another. Early proponents of semantic decomposition (Katz & Fodor, 1963; Katz, 1972; Gruber, 1976; Lakoff, 1970; Jackendoff, 1972; Schank, 1972; Miller and Johnson-Laird, 1976; and others), whether in a generative or an interpretative framework, argued for the existence of a finite set of universal semantic-conceptual components (or primes, or primitives, or atomic predicates, or, in the case of nouns, markers) into which all lexical items could be exhaustively decomposed. Among the examples one finds discussed in the literature, most have been English verbs. The best known example of the decomposition of a verb is probably McCawley's (1968) analysis of *kill* into CAUSE TO BECOME NOT ALIVE, which has been much discussed and criticized (Fodor, 1970; Shibatani, 1972; and others).

Although semantic decomposition has been judged by some to be an inadequate theory of semantic representation (Chomsky, 1972, and others), more recent approaches have taken a similar path to semantic analysis. Both Jackendoff (1983) and Talmy (1985) have proposed an analysis of verbs in terms of such conceptual categories as EVENT, STATE, ACTION, PATH, MANNER, PLACE, etc. For example, Talmy analyzes the verb *roll* as being a lexicalized conflation of MOVE and MANNER. Both analyses share the assumption of a limited inventory including components or categories that can be expressed not only by verbs, but also by nouns and by operators like NEG.

Relational semantic analysis differs from semantic decomposition primarily in taking lexical items, rather than hypothetically irreducible meaning atoms, as the smallest unit of analysis. Thus, relational analysis has the advantage that its units can be thought of as entries in speakers' mental dictionaries. However, the relational analysis adopted in WordNet shares some aspects of decomposition. Although WordNet does not explicitly recognize conceptual components, some components are reflected in the nature of the semantic relations linking verbs to each other. For example, one of generative semantics' important subpredicates, CAUSE, has the status of a semantic relation in WordNet—a relation that links such verb pairs as *teach-learn* and *show-see*. This relation also distinguishes systematically between the causative (transitive) and the anticausative (intransitive) senses of certain verb classes, including *break*, *rot* and *move*.

Components like NEG and PATH do not play an overt role in WordNet. But NEG is clearly implicit in the opposition relation that holds between contradictory verbs like *live* and *die*, or *succeed* and *fail*, and between gradables like *like* and *dislike*. And a semantic PATH component is clearly part of the manner relation that links verb concepts like {*move*, *travel*} and *soar*. Other features, such as Talmy's MANNER, are also part of the semantics of WordNet's manner relation (troponymy) linking basic verbs like *eat* and *communicate* to other verbs denoting particular elaborations of the base verb, such as *gobble* and *telex*.

Proponents of semantic decomposition have also argued for the existence of subpredicates, corresponding to abstract verbal concepts, in many lexicalized verbs. Some abstract semantic predicates such as MOVE or GO are argued to be the basic components of most verbs from a wide variety of different semantic fields (Gruber, 1976;

Jackendoff, 1972, 1976). Gruber's Location Hypothesis argues that all events and states can be analyzed as more or less abstract spatial locations and motions. Similarly, Dowty (1979) analyzes all English verbs with the exception of statives as verbs of change, and posits as part of their meaning the semantic subpredicate CHANGE. Thus, Gruber analyzes verbs of giving as an abstract motion undergone by the object that is given; Dowty sees giving as a change of possession. Such decompositions of verb semantics are defensible on an abstract plane, but it is not clear that they reflect the way in which people's memory of verbs is structured; there is no evidence that speakers store verbs of giving together with the concept of (abstract) motion or change of location. Pinker (1989:101) claims that speakers of English decompose verbs into such semantic subpredicates as CAUSE, GO, BE, and PATH, which enables them to predict the verbs' idiosyncratic syntactic behavior. Such an analysis may well be part of speakers' linguistic competence, but there is no indication that it serves to organize the mental lexicon. And there is no evidence that verbs with a complex composition take longer to use or understand.

The subpredicates of lexical decomposition are accorded the same status as other verbs in WordNet. Because verbs like {*move*, *go*} and *change* refer to very basic concepts, they constitute the root verbs, or topmost "unique beginners," heading two semantic fields. The verbs making up these fields are linked to the root verbs via semantic relations. In other words, WordNet characterizes the verbs of change not by the presence of a semantic component CHANGE, but rather by a semantic relation to the verb *change*. This distinction may appear subtle; it hinges on the formulation of the semantic relations that are coded in WordNet.

Componential analysis could be viewed in terms of entailment, in that a verb V_1 that is a component of another verb V_2 must be entailed by V_1 . This idea forms the basis of Carnap's (1947) theory of meaning postulates. Rather than attempting to provide an exhaustive breakdown of words into components, meaning postulates state inferential rules between sentences based on the semantic composition of the words in these sentences. A well-known example is the relation between the sentence *John is a bachelor* and the sentences *John is a male* / *John is an adult* / *John is unmarried*. These entailment relations reflect the fact that the lexical composition of *bachelor* includes the components MALE, ADULT, and UNMARRIED. The meaning of *bachelor* can be expressed in terms of its position in a taxonomic hierarchy, that is, a *bachelor* is a kind of a *man*, which is a kind of a *person*. *Bachelor*, therefore, could be said to inherit all the semantic components of its superordinates, such as HUMAN. Consequently, the sentence *John is a bachelor* also entails the sentence *John is human*. (Katz, 1972, expresses a similar idea by postulating lexical redundancy rules concerning the co-occurrence of semantic components.) Carnap's meaning postulates also predict an entailment relation between such verbs as *kill* and *die*. The entailment here reflects the fact that *die* and *kill*, which is decomposed into CAUSE TO DIE, share the semantic component DIE.

Although WordNet avoids semantic decomposition in favor of a relational analysis, the semantic relations among verbs in WordNet all interact with entailment.

Lexical Entailment

The principle of lexical inheritance can be said to underlie the semantic relations between nouns, and bipolar oppositions serve to organize the adjectives. Similarly, the different relations that organize the verbs can be cast in terms of one overarching principle, lexical entailment.

In logic, entailment, or strict implication, is properly defined for propositions; a proposition *P* entails a proposition *Q* if and only if there is no conceivable state of affairs that could make *P* true and *Q* false. Entailment is a semantic relation because it involves reference to the states of affairs that *P* and *Q* represent. The term will be generalized here to refer to the relation between two verbs *V*₁ and *V*₂ that holds when the sentence *Someone V*₁ logically entails the sentence *Someone V*₂; this use of entailment can be called lexical entailment. Thus, for example, *snore* lexically entails *sleep* because the sentence *He is snoring* entails *He is sleeping*; the second sentence necessarily holds if the first one does.

Lexical entailment is a unilateral relation: if a verb *V*₁ entails another verb *V*₂, then it cannot be that case that *V*₂ entails *V*₁. The exception is that where two verbs can be said to be mutually entailing, they must also be synonyms, that is, they must have the same sense. For example, one might say both that *The Germans beat the Argentinians* entails *The Germans defeated the Argentinians*, and that *The Germans defeated the Argentinians* entails *The Germans beat the Argentinians*. However, we find such statements rather unnatural. Negation reverses the direction of entailment: *not sleeping* entails *not snoring*, but *not snoring* does not entail *not sleeping*. The converse of entailment is contradiction: If the sentence *He is snoring* entails *He is sleeping*, then *He is snoring* also contradicts the sentence *He is not sleeping* (Kempson, 1977).

The entailment relation between verbs resembles meronymy between nouns, but meronymy is better suited to nouns than to verbs. To begin with, in order for sentences based on the formula *An x is a part of a y* to be acceptable, both *x* and *y* must be nouns. It might seem that using the nominalizing gerundive form of the verbs would convert them into nouns, and as nouns the HASA relation should apply. For example, Rips and Conrad (1989) obtained consistent results when they asked subjects to judge questions like *Is thinking a part of planning?* vs. *Is planning a part of thinking?* But this change in syntactic category does not overcome fundamental meaning differences between nouns and verbs. Fellbaum and Miller (1990) argue that, first, verbs cannot be taken apart in the same way as nouns, because the parts of verbs are not analogous to the parts of nouns. Most nouns and noun parts have distinct, delimited referents. The referents of verbs, on the other hand, do not have the kind of distinct parts that characterize objects, groups, or substances. Componential analyses have shown that verbs cannot be broken up into referents denoted solely by verbs. And, second, the relations among parts of verbs differ from those found among noun parts. Any acceptable statement about part-relations among verbs always involves the temporal relation between the activities that the two verbs denote. One activity or event is part of another activity or event only when it is part of, or a stage in, its temporal realization.

It is true that some activities can be broken down into sequentially ordered subactivities. For the most part, these are complex activities that are said to be mentally represented as scripts (Schank and Abelson, 1977). They tend not to be lexicalized in English: *eat at a restaurant*, *clean an engine*, *get a medical check-up*, etc. The analysis into lexicalized sub-activities that is possible for these verb phrases is, however, not available for the majority of simple verbs in English. Yet people will also accept part-whole statements involving verb pairs like *drive-ride* and *snore-sleep*. The reason lies in the kinds of entailment that hold between the verbs.

Consider the relation between the verbs *ride* and *drive*. Although neither activity is a discrete part of the other, the two are connected in that when you drive a vehicle, you necessarily also ride in it. The relation between the two activities denoted by these verbs is different from that holding between an activity like *get a medical check-up* and its temporally sequential sub-activities, like *visit (the doctor)* and *undress*. *Riding* and *driving* are carried on simultaneously. Yet most people accept *Riding is a part of driving* and reject *Driving is a part of riding*, even though neither activity can be considered a subactivity of the other.

Consider also the relations among the activities denoted by the verbs *snore*, *dream*, and *sleep*. Snoring or dreaming can be a part of sleeping, in the sense that the two activities are, at least partially, temporally co-extensive: the time that you spend snoring or dreaming is a proper part of the time you spend sleeping. And it is true that when you stop sleeping you also necessarily stop snoring or dreaming.

The differences between pairs like *drive* and *ride* and *snore* and *sleep* are due to the temporal relations between the members of each pair. The activities can be simultaneous (as with *drive* and *ride*) or one can include the other (as with *snore* and *sleep*). For both pairs, engaging in one activity necessitates engaging in the other activity. Therefore, the first activity in each pair entails the second.

The two semantic relations subsumed under lexical entailment that we have considered so far share the feature of temporal inclusion. That is to say, the sets of verbs related by entailment have in common that one member temporally includes the other. A verb V_1 will be said to include a verb V_2 if there is some stretch of time during which the activities denoted by the two verbs co-occur, but no time during which V_2 occurs and V_1 does not. If there is a time during which V_1 occurs but V_2 does not, V_1 will be said to properly include V_2 .

Temporal inclusion may go in either direction. Verb pairs like *buy* and *pay* differ from those like *snore* and *sleep* in that whereas *snore* entails *sleep* and is properly included by it, *buy* entails *pay* but properly includes it. That is to say, either the entailing or the entailed verb may properly include the other. In the case of *snore-sleep*, it is the entailed verb *sleep* that properly includes the entailing verb *snore*, whereas in the pair *buy-pay*, the entailing verb *buy* properly includes the entailed verb *pay*.

Our analysis so far yields a simple generalization: if V_1 entails V_2 , and if a temporal inclusion relation holds between them, then people will accept a part-whole statement relating V_2 and V_1 .

Hyponymy Among Verbs

The sentence frame used to test hyponymy between nouns, *An x is a y*, is not suitable for verbs, because it requires that *x* and *y* be nouns: *to amble is a kind of to walk* is not a felicitous sentence. Even when this formula is used with verbs in the gerundive form, there is a noticeable difference between nouns and verbs. Although people are quite comfortable with statements like *A horse is an animal* or *A spade is a garden tool*, they are likely to reject such statements as *Ambling is walking* or *Mumbling is talking*, where the superordinate is not accompanied by some qualification. The semantic distinction between two verbs is different from the features that distinguish two nouns in a hyponymic relation.

An examination of 'verb hyponyms' and their superordinates shows that lexicalization involves many kinds of semantic elaborations across different semantic fields. For example, Talmy's (1985) analysis of motion verbs treats them as confluences of *move* and such semantic components as MANNER and CAUSE, exemplified by *slide* and *pull*, respectively. To these components could be added SPEED (encoded in *run*, *stroll*) or the CONVEYANCE of displacement (*bus*, *truck*, *bike*). Similarly, English verbs denoting different kinds of *hitting* express the DEGREE OF FORCE used by the agent (*chop*, *slam*, *whack*, *swat*, *rap*, *tap*, *peck*, etc.). Some verbs refer to different degrees of INTENSITY of the action or state (*drowse*, *doze*, *sleep*).

Since the aim is to study relations between verbs, rather than between the building blocks that make them up, the many different kinds of elaborations that distinguish a 'verb hyponym' from its superordinate have been merged into a manner relation that Fellbaum and Miller (1990) have dubbed troponymy (from the Greek *tropos*, manner or fashion). The troponymy relation between two verbs can be expressed by the formula *To V₁ is to V₂ in some particular manner*. 'Manner' is interpreted here more loosely than in Talmy's work, for example, and troponyms can be related to their superordinates along many semantic dimensions. Subsets of particular kinds of manners tend to cluster within a given semantic field. Among competition verbs, for example, many troponyms are confluences of the basic verb *fight* with nouns denoting the occasion for, or form of, the fight: *battle*, *war*, *tourney*, *joust*, *duel*, *feud*, etc. Troponyms of communication verbs often encode the speaker's INTENTION or motivation for communicating, as in *examine*, *confess*, or *preach*, or the MEDIUM of communication: *fax*, *e-mail*, *phone*, *telex*.

Troponymy and Entailment

Troponymy is a particular kind of entailment, in that every troponym *V₁* of a more general verb *V₂* also entails *V₂*. Consider the pair *limp-walk*. The verbs in this pair are related by troponymy: *to limp* is also *to walk in a certain manner*; *limp* is a troponym of *walk*. The verbs are also in an entailment relation: the statement *He is limping* entails *He is walking*, and walking can be said to be a part of limping. Unlike the activities denoted by *snore* and *sleep*, or *buy* and *pay*, the activities referred to by a troponym and its more general superordinate are always temporally co-extensive, in that one must necessarily be walking every instant that one is limping. Troponymy therefore represents a special case of entailment: pairs that are always temporally co-extensive and are related by entailment.

In contrast with pairs like *limp-walk*, a verb like *snore* entails and is included in *sleep*, but is not a troponym of *sleep*; *get a medical check-up* entails and includes *visit the doctor*, but is not a troponym of *visit the doctor*; and *buy* entails *pay*, but is not a troponym of *pay*. The verbs in these pairs are related only by entailment and proper temporal inclusion. The important generalization here is that verbs related by entailment and proper temporal inclusion cannot be related by troponymy. For two verbs to be related by troponymy, the activities they denote must be temporally co-extensive. One can sleep before or after snoring, buying includes activities other than paying, and visiting the doctor is not temporally co-extensive with getting a medical check-up, so none of these pairs are related by troponymy.

The two categories of lexical entailment that have been distinguished so far are related diagrammatically in Figure 1.

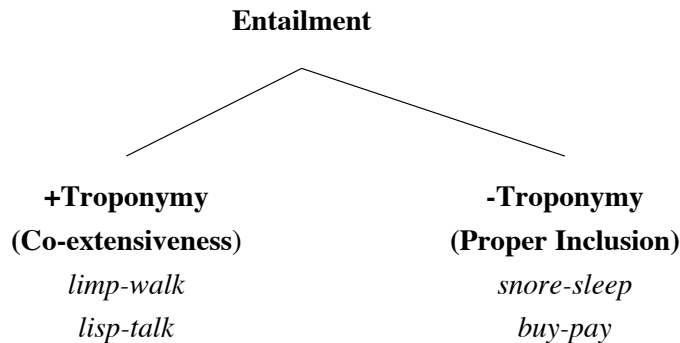


Figure 1. Two kinds of entailment with temporal inclusion

Verb Taxonomies

In trying to construct verb taxonomies using the troponymy relation, it became apparent that verbs cannot easily be arranged into the kind of tree structures onto which nouns are mapped. First, within a single semantic field it is frequently the case that not all verbs can be grouped under a single unique beginner; some semantic fields must be represented by several independent trees. Motion verbs, for example, have two top nodes, {*move, make a movement*}, and {*move, travel*}. Verbs of possession can be traced up to the three verbs {*give, transfer*}, {*take, receive*}, and {*have, hold*}; for the most part, their troponyms encode ways in which society has ritualized the transfer of possessions: *bequeath, donate, inherit, usurp, own, stock*, etc. The semantic field containing verbs of bodily care and functions consists of a number of independent hierarchies that form a coherent semantic field by virtue of the fact that most of the verbs (*wash, comb, shampoo, make up; ache, atrophy*) select for the same kinds of noun arguments (body parts). The communication verbs are headed by the verb *communicate* but immediately divide into verbs of verbal and non-verbal (gestural) communication; the former divide further into actions denoting the communication of spoken vs. written

language.

Verb hierarchies tend to have a more shallow, bushy structure than nouns; in few cases does the number of hierarchical levels exceed four. Moreover, virtually every verb taxonomy shows what might be called a bulge, that is to say, a level far more richly lexicalized than the other levels in the same hierarchy. Call this layer *L0*, the layer above it *L+1*, and the layer below *L-1*. Certain parallels can be drawn between *L0* and what has been called the basic level in noun hierarchies (Rosch et al., 1976). Not only do most of the verbs in a hierarchy cluster at *L0*, but the troponymy relation between these verbs and their superordinate is semantically richer than between verbs on other levels. Consider, for example, the taxonomy arising from the *L+1* verb *walk*: the superordinate of *walk*, on level *L+2*, is {*move, travel*}; troponyms of *walk*, on level *L0*, are *march, strut, traipse, amble, mosey, slouch*, etc. Although a statement relating *L+1* to *L+2*—*To walk is to move in some manner*—is perfectly acceptable, statements relating *L0* to *L+1*—*To march/strut/traipse/amble . . . is to walk in some manner*—seem more felicitous; these verbs elaborate the concept of *walking* in distinct ways, yet the features of *walking* are still clearly present. *Walk*, on the other hand, seems semantically more remote from its superordinate, *move*.

An alternative way to think about verb taxonomies—one that reflects the prominence of the two levels, *L+1* and *L0*—is in terms of a radial structure, or cluster, with unelaborated *L+1* verbs like *walk, talk, hit*, and *fight* in the center and their troponyms *march, strut; lisp, babble; tap, slam; battle, joust*, etc., clustered around them.

In most hierarchies, *L-1*, the level below the most richly lexicalized one, has few members. For the most part, they tend not to be independently lexicalized, but are compounded from their superordinate verb and a noun or noun phrase. Examples are *goose-step*, a troponym of *march* from the *walk* hierarchy; and *spoonfeed, forcefeed, bottle-feed, breastfeed*, troponyms of {*feed, cause to eat*} from the *ingest* hierarchy among the consumption verbs.

As one descends in a verb hierarchy, the variety of nouns that the verbs on a given level can take as potential arguments decreases. This seems to be a function of the increasing elaboration and meaning specificity of the verb. Thus, *walk* can take a subject referring either to a person or an animal; most troponyms of *walk*, however, are restricted to human subjects. And *goose-stepping* is usually, though not necessarily, done by soldiers; this verb rarely takes *children* or *old people* as arguments. On the other hand, {*move, travel*} can take not only person and animal subjects, but also vehicles, or objects moved by external forces. Similarly, figures or pictures can *communicate* and *talk*; they can even *deceive* or *lie*, but they cannot *fib* or *perjure themselves*, as only human speakers can. A piece of news may *hit, touch* or even *grab* you, but it cannot *punch, stroke* or *collar* you; only people can be agents of these verbs.

Opposition Relations between Verbs

There is evidence that opposition relations are psychologically salient not only for adjectives, but also for verbs. For example, in the course of teaching foreign languages the author has experienced that students, when given only one member of an antonymous

or opposed verb pair, will insist on being told the other member; they believe that it is easier to learn semantically opposed words together. Fellbaum and Chaffin (1990), in an analogy task involving different semantic relations between verbs, asked subjects to generate verbs whose relation to the stimulus matched that of a given pair; they found that subjects were most successful in completing analogies that involved an opposition relation. Moreover, analogies based on opposition relations took the least time to complete. In building the database for verbs, it was found that after synonymy and troponymy, opposition is the most frequently coded semantic relation.

The semantics of opposition relations among verbs is complex. As in the case of adjectives, much of the opposition among verbs is based on the morphological markedness of one member of an opposed pair, as in the pairs *tie/untie* and *appear/disappear*/fR.

Like the semantically similar adjective pairs *weighty/weightless* and *heavy/light*, there are pairs like *fall/rise* and *ascend/descend* that seem identical in meaning, yet are distinguished by the way their members pick out their direct antonyms: *rise/descend* and *ascend/fall* are conceptually opposed, but are not direct antonyms.

Many deadjectival verbs formed with a suffix such as *-en* or *-ify* inherit opposition relations from their root adjectives: *lengthen/shorten*, *strengthen/weaken*, *prettify/uglify*, for example. These are, for the most part, verbs of change and would decompose into BECOME + adjective or MAKE + adjective. As in the case of the corresponding adjectives, these are direct antonyms. Synonyms of these verbs, when they exist, are generally of Latin or Greek origin and tend to be more constrained in the range of their potential arguments, that is to say, they are usually reserved for more specialized uses. Thus, *fortify* is a synonym of *strengthen*, but its opposition relation to *weaken* is conceptual: *fortify/weaken* are indirect antonyms. In short, deadjectival verbs can be represented by the same configuration that Gross and Miller (this volume) describe for adjectives.

As in the case of adjectives, a variety of negative morphological markers attach to verbs to form their respective opposing members. Examples are *tie/untie*, *approve/disapprove*, and *bone/debone*. The semantics of these morphological oppositions is not simple negation. *To untie* is a kind of *undoing*, and the sense of this verb is one of reversing an action; it does not mean *to not tie*. A pair of verbs like *approve/disapprove* are gradables: the two lexicalized terms are points on a scale (of approval, in this case). Gradable verbs, like gradable adjectives, can be modified by degree adverbs, such as *quite*, *rather*, etc. Perhaps the most striking example illustrating that negative morphology is not simple negation is seen in such pairs as *bone/debone* where, despite the lexical opposition induced by the presence of a negative prefix, there is no semantic opposition at all—both verbs refer to the same activity of removing the bones of an animal (Horn, 1989). In some pairs, the marked member cannot be inferred simply from the morphological marker because the opposition derives from the prefixes themselves: *emigrate/immigrate*, *exhale/inhale*, *predate/postdate*.

Other pairs whose members seem to be direct antonyms are *rise/fall* and *walk/run*. Members of these pairs are associated with each other rather than with verbs that are

synonyms of their respective opposites and that express the same concept as that opposite. These pairs are illustrative of an opposition relation that is found quite systematically between co-troponyms (troponyms of the same superordinate verb). For example, the motion verbs *rise* and *fall* both conflate the superordinate *move* with a semantic component denoting the direction of motion; they constitute an opposing pair because the direction of motion, upward or downward, is opposed. Similarly, the opposition between *walk* and *run*, two co-troponyms of {*move*, *travel*}, is due to the opposing manners (slow or fast, respectively) that distinguish each troponym from its superordinate. And the opposition of *nibble* and *gorge*, co-troponyms of *eat*, derives from the opposition between the quantities eaten. Similarly, *breastfeed* and *bottlefeed* are opposites because they refer to two opposite manners of feeding an infant.

Still other pairs illustrate the variety and complexity of verb opposition. Some pairs, called converses, are opposites that are associated with no common superordinate or entailed verb: *give/take*, *buy/sell*, *lend/borrow*, *teach/learn*, etc. They have in common that they occur within the same semantic field: they refer to the same activity, but from the viewpoint of different participants. This fact would lead one to surmise that their strong lexical association is probably due to their frequent co-occurrence in usage.

Most antonymous verbs are stative or change-of-state verbs that can be expressed in terms of attributes. There are many opposition relations among stative verbs: *live/die*, *exclude/include*, *differ/equal*, *wake/sleep*. Opposition relations are also frequent among change verbs. Virtually no other relation (other than synonymy) holds these verbs together. Thus, the organization of this suburb of the lexicon is flat rather than hierarchical—there are no superordinates (except the generic *change* and *be* or *have*), and virtually no troponyms. Change verbs and stative verbs thus have a structure resembling that of the adjectives, with only synonymy and opposition relations.

Opposition and Entailment

Many verb pairs in an opposition relation also share an entailed verb. For example, both *hit* and *miss* entail *aim*, because one must necessarily aim in order to hit or miss. In contrast to the kinds of entailment discussed earlier, these verbs are not related by temporal inclusion. The activities denoted by *hit* (or *miss*) and *aim* occur in a sequential order: in order to either hit or miss, one must have aimed first; aiming is a precondition for both hitting and missing. The relation between the entailing and the entailed verbs here is one of backward presupposition, where the activity denoted by the entailed verb always precedes the activity denoted by the entailing verb in time. Other examples are *fail* and *succeed*, which both entail *try*; and *win* and *lose*, both entailing *play* or *gamble*.

Entailment via backward presupposition also holds between certain verb pairs related by a result or purpose relation, such as *fatten-feed*.

A verb V_1 that is entailed by another verb V_2 via backward presupposition cannot be said to be a part of V_2 . Part-whole statements between verbs are possible only when a temporal inclusion relation holds between these verbs.

The set of verbs related by entailment that we have considered so far can be classified exhaustively into two mutually exclusive categories on the basis of temporal

inclusion (see Fig.2).

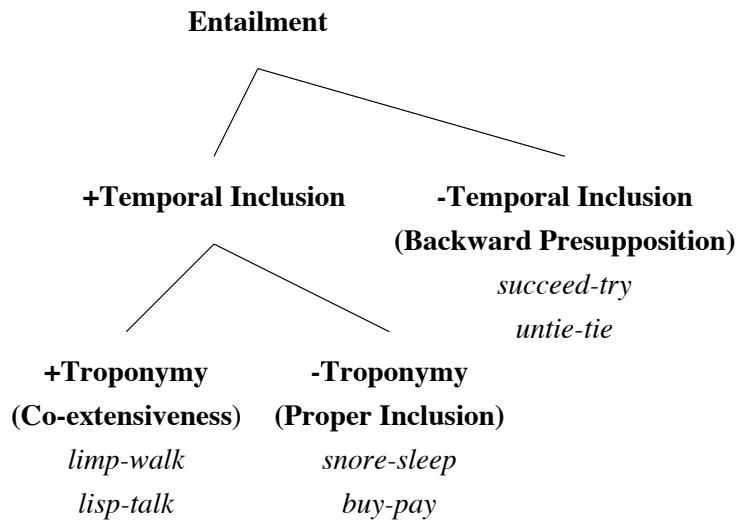


Figure 2. Three kinds of entailment

Opposing verbs like *fail* and *succeed* tend to be contradictories, and, like contradictory adjectives, they do not tolerate degree adverbs. Some opposition relations interact with the entailment relation in a systematic way. Cruse (1986) distinguishes an opposition relation that holds between verb pairs like *damage* and *repair*, and *remove* and *replace*. One member of these pairs, Cruse states, constitutes a “restitutive.” This kind of opposition also always includes entailment, in that the restitutive verb always presupposes what one might call the “deconstructive” one. Many reversive *un-* or *de-* verbs also presuppose their unprefixd, opposed member: in order to *untie* or *unscrew* something, someone must have *tied* or *screwed* it first. Again, these accompanying entailment relations are not marked in WordNet; only the more salient opposition relation is entered.

The Causal Relation

The causative relation picks out two verb concepts, one causative (like *give*), the other what might be called the “resultative” (like *have*). In contrast to the other relations coded in WordNet, the subject of the causative verb usually has a referent that is distinct from the subject of the resultative; the subject of the resultative must be an object of the causative verb, which is therefore necessarily transitive. The causative member of the pair may have its own lexicalization, distinct from the resultative, as in the pair *show* and *see*; sometimes, the members of such a pair differ only by a small variation in their common stem, as in the case of *fell-fall* and *raise-rise*. Although many languages have a means to express causation, not all languages lexicalize the causative member

independently; causation is often marked by a morpheme reserved for this function. English does not have many lexicalized causative-resultative pairs, such as *show-see*; it has an analytic, or periphrastic, causative, formed with *cause to/make/let/have/get to*, that is used productively.

It has frequently been pointed out that a periphrastic causative is not semantically equivalent to a lexicalized causative (Fodor, 1970; Shibatani, 1972; and others), but refers to a more indirect kind of causation than the direct, lexicalized form. *Kill* and *cause to die* usually cannot be used interchangeably to refer to the same action, and so are not strictly speaking synonymous expressions of the same concept. For example, Chomsky (1977) notes that you can cause someone to die by having him drive across the country with a pathological murderer, but your action could not properly be called killing. For the purposes of WordNet, such pragmatic considerations have been disregarded.

WordNet recognizes only lexicalized causative-resultative pairs. The synonyms of the members of such a pair inherit the Cause relation, indicating that this relation holds between the entire concept rather than between individual word forms only: the synonyms {*teach, instruct, educate*}, for example, are all causatives of the concept {*learn, acquire knowledge*}. However, unlike entailment, the causation relation is not inherited by the troponyms of each of these concepts: *spoonfeed, indoctrinate*, and *tutor* do not necessarily cause the student to *cram, stuff*, or *memorize*.

Causative verbs have the sense of *cause to be/become/happen/have* or *cause to do*. That is to say, they relate transitive verbs to either states or actions. For example, *give* and *teach* are related via causation to the statives *have* and *know*; *raise* and *feed* are related to the events or actions referred to by *rise* and *eat*. In both cases, causation can be seen as a kind of change. Many verbs clearly have the semantics of such a causative change, but they do not have lexicalized resultatives. The *amuse* and *annoy* subgroup of the psych verbs all refer to causing the experiencer to have an emotion, but only one such causative concept, {*frighten, scare*}, has a lexicalized resultative, {*fear, dread*}.

There are many verbs in English that have both a causative and an anticausative usage. Most of them cluster in the file containing the verbs of change, where many verbs alternate between a transitive causative form and an intransitive anticausative (or unaccusative, or inchoative) form. Here, the surface form of the causative and the anticausative verbs are identical. Examples are the verbs *whiten, grow, break*, and *shrink*. Most anticausative verbs imply either an animate agent or an inanimate cause (*The glass door broke-The storm/The children broke the glass door*). A few verbs are compatible only with an inanimate cause: *Johnnie's teeth rotted—All that candy rotted Johnnie's teeth*, is acceptable, but *His mother rotted Johnnie's teeth* is not.

The causative relation also shows up systematically among the motion verbs: *bounce, roll, blow*, etc., alternate between a causative and an anticausative usage (*She blew a soap bubble in his face vs. The soap bubble blew in his face*). While the causative variants of these verbs usually require an inanimate object, some unergative verbs like *run, jump, gallop, walk, race*, which select for an animate agent, can also have a causative reading, as in the sentences *He raced the horse past the barn* and *The father*

walked his son to school. (See Levin, 1985, and Pinker, 1989, for a conceptual analysis of these verbs.)

Causation and Entailment

Carter (1976) notes that causation is a specific kind of entailment: if V_1 necessarily causes V_2 , then V_1 also entails V_2 . He cites the entailment relation between verb pairs like *expel* and *leave*, or *bequeath* and *own*, where the entailing verb denotes the causation of the state or activity referred to by the entailed verb. Like the backward presupposition relation that holds between verbs like *fail/succeed* and *try*, the entailment between verbs like *bequeath* and *own* is characterized by the absence of temporal inclusion.

The causation relation is unidirectional: although *giving* something to somebody causes the recipient to *have* it, for someone to *have* something does not entail that he was *given* it. Similarly, *feeding* somebody causes that person to *eat*, but somebody's *eating* does not entail that someone *feeds* the eater. Except that when the subject of *eat* is not a potentially independent agent, such as a baby or a confined animal, then *eating* does entail the causative act of *feeding*. The direction of the entailment may therefore be reversed in such cases, depending on the semantic features of the verb's subject. But because the entailment depends on specific features of the subject, it can no longer be said to be lexical, that is, it is no longer a relation between two verbs only.

We have now distinguished four different kinds of lexical entailment that systematically interact with the semantic relations we code in WordNet. These four kinds of entailment are related in Figure 3.

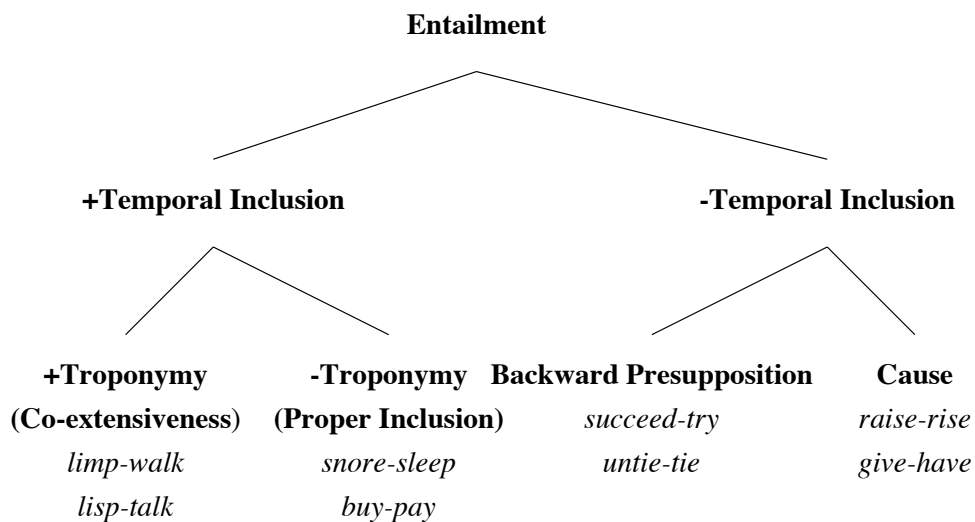


Figure 3. Four kinds of entailment relations among verbs

Syntactic Properties and Semantic Relations

In recent years the lexicon has gained increasing attention from linguists. Verbs in particular have been the subject of much research in pursuit of a theory of lexical knowledge. The work of Levin (1985, 1989) and others has focused on properties of verbs as lexical items that combine with noun arguments to form sentences. This research analyzes the constraints on verbs' argument-taking properties in terms of their semantic makeup, based on the assumption that the distinctive syntactic behavior of verbs and verb classes arises from their semantic components. Pinker (1989) undertakes a fine-grained semantic analysis of verbs that participate in syntactic alternations, claiming that children can discern subtle differences that enable them to distinguish semantically-based verb classes with certain syntactic properties. Gleitman (1990) asserts that children exploit the syntactic-semantic regularities of verbs to infer their meanings on the basis of their syntactic properties.

WordNet was designed to model lexical memory rather than represent lexical knowledge, so it excludes much of a speaker's knowledge about both semantic and syntactic properties of verbs. There is no evidence that the syntactic behavior of verbs (or any other lexical category) serves to organize lexical memory. But there is a substantial body of research (cited in Levin, 1989) showing undeniable correlations between a verb's semantic make-up and its syntax, and the possible implications for children's acquisition of lexical knowledge (Pinker, 1989; Gleitman, 1990).

To cover at least the most important syntactic aspects of verbs, therefore, WordNet includes for each verb synset one or several sentence frames, which specify the subcategorization features of the verbs in the synset by indicating the kinds of sentences they can occur in. This information permits one quickly to search among the verbs for the kinds of semantic-syntactic regularities studied by Levin and others. One can either search for all the synsets that share one or more sentence frames in common and compare their semantic properties; or one can start with a number of semantically similar verb synsets and see whether they exhibit the same syntactic properties. An exploration of the syntactic properties of co-troponyms occasionally provides the basis for distinguishing semantic subgroups of troponyms.

As a case in point, consider verbs like *fabricate* and *compose*, which are members of the creation verb class. Many creation verbs participate in a syntactic alternation that Levin (1989) terms the Material/Product alternation, illustrated by the following examples:

She wove a rug from the black sheep's wool
She wove the black sheep's wool into a rug
They molded a head from the clay
They molded the clay into a head

Some verbs, like *fabricate* and *compose*, which also share membership in the class of creation verbs, do not participate in this syntactic alternation, despite their semantic similarity to verbs like *weave* and *mold*:

The reporter fabricated a story out of the girl's account

*The reporter fabricated the girl's account into a story

She composed a quartet out of the old folk song

*She composed the old folk song into a quartet

In discussing these verbs, Fellbaum and Kegl (1988) point out that the data suggest a need for a fine-grained sub-classification of creation verbs that distinguishes a class of verbs referring to acts of mental creation (such as *fabricate* and *compose*) from verbs denoting the creation from raw materials (such as *weave* and *mold*). Such a distinction would account for the systematic difference among the verbs in most cases. Thus, Levin (1989) distinguishes these verbs in terms of membership in one of two classes: the BUILD class, which comprises verbs like *bake*, and the CREATE class constituted by such verbs as *compose* and *fabricate*. However, English does not have a lexicalized generic verb denoting the concepts of *create from raw material* and *create mentally*, which would make it possible to capture this generalization by means of different superordinates whose troponyms differ syntactically. But the observation can be formulated in terms of differences in the manner relations that link verbs like *mold* on the one hand, and verbs like *compose* on the other hand, to their common superordinate *create*. This example demonstrates how syntactic differences between apparently similar verbs can be cast in terms of the particular way that the meanings of words are represented in WordNet.

Viewing verbs in terms of semantic relations can also provide clues to an understanding of the syntactic behavior of verbs. Fellbaum and Kegl (1989) studied a class of English verbs that participate in the following transitive-intransitive alternation:

Mary ate a bag of pretzels

Mary ate

Previous analyses of these verbs have explained the alternation in terms of discourse control (Fillmore, 1986) or aspect (Mittwoch, 1982). However, an analysis of the troponyms of the verb *eat* showed that they fall into two syntactic classes: those that must always be used transitively, and those that are always intransitive. The first class includes the verbs *gobble*, *guzzle*, *gulp*, and *devour*; the second class includes verbs like *dine*, *graze*, *nosh*, and *snack*. Fellbaum and Kegl suggest that this syntactic difference is not just a transitivity alternation characteristic of a single verb, but is semantically motivated. They show that English has two verbs *eat*, and that each verb occupies a different position in the network, that is to say, each verb is part of a different taxonomy. Intransitive *eat* has the sense of *eat a meal*. In some troponyms of this verb, such as the denominals *dine*, *breakfast*, *picnic*, and *feast*, the verb *eat* has become conflated with hyponyms of the noun *meal*. These verbs are intransitive because they are all lexicalizations of the verb *eat* that means *eat a meal*. Other intransitive troponyms of this verb are *munch*, *nosh*, and *graze*. Although these verbs are not conflations of *eat* and a noun, they are semantically related in that they refer to eating informal kinds of meals or repasts. By contrast, the transitive verb *eat* has the sense of *ingest in some manner*, and its troponyms all refer to a specific manner of eating: *gobble*, *gulp*, *devour*, etc.³ Thus,

³ Kenneth Hale (personal communication) informs us that two *eat* verbs having this semantic distinction are found cross-linguistically.

the semantics of the troponyms in each case provide a classification in terms of two distinct hierarchies matching the syntactic distinction between the two verb groups.

Summary

Different semantic groups of verbs have distinct structures. Some parts can be cast into a taxonomic framework by means of the troponymy relation; this is generally true for verbs of creation, communication, competition, contact, motion, and consumption. The troponymy relation covers a number of different manner relations, one or more of which tend to cluster in specific semantic domains. In some semantic domains, distinct patterns of lexicalization can be discerned; often a base verb conflates with nouns from the corresponding semantic domain. In such verb hierarchies, which tend to be much flatter than noun hierarchies, one level can be distinguished that is more richly lexicalized than the other levels. Stative verbs and verbs of change exhibit an entirely different structure: they tend to be organized in terms of opposition and synonymy relations and they can be mapped into the bipolar clusters that characterize the adjectives.

Comments on Specific Verb Clusters

This section lists particular properties of the several verb files.

1. *Verbs of Bodily Functions and Care*. This relatively small file of approximately 275 synsets contains many verbs referring to actions or events that are not under the control of the argument that functions as their subject (*sweat, shiver, faint, burp, ache, tire, sleep, freeze*, etc.). For this reason these verbs, which are mostly intransitives, have been called unaccusatives, that is, their subjects are thought to be underlyingly objects. Some verbs, like *snort* and *wink*, that have a basic sense referring to an involuntary action, acquire a sense of communication when the action is intended and controlled by the agent. A number of body verbs referring to grooming activities such as *wash*, and *dress* have a reflexive reading in their intransitive form, and are transitive when the action is performed on someone other than the agent, or on a specific body part of the agent.

2. *Verbs of Change*. The verbs of change constitute one of the largest verb files in WordNet (about 750 synsets), owing in part to the fact that the concept of *change* is flexible enough to accommodate verbs whose semantic description makes them unfit for any other semantically coherent group. Dowty (1979) analyzes all verbs as ultimately being composable as stative predicates, which, by means of aspectual connectives and operators such as DO, yield other verb classes. With the exception of the statives, Dowty additionally assigns to all verbs the operator BECOME as part of their lexical make-up. His analysis shows that all (non-stative) verbs can be classified as verbs of change, either as intransitives (stative predicates plus the operator BECOME) or as transitives (with the additional operator DO). An analysis like Dowty's is based on the decomposition of verbs, rather than on a relational analysis of the kind WordNet has undertaken, but it shows that, given the abstract concept of CHANGE, all verbs can be derived from it. In WordNet, this concept has been broken down into several superordinate verbs of change: {*change, alter, vary, modify*}, {*change1, change state*}, {*change2, change by reversal*,

turn, revert}, {*change integrity*}, {*change shape*}, and {*change3, adjust, conform, adapt*}. Most verbs in the change file are derived from these verbs via troponymy. Thus, WordNet characterizes the verbs of change not by the presence of a semantic component CHANGE but by a semantic relation to one of these more specific concepts of change.

The fact that the change file is so large is also due, in part, to the fact that English has productive morphological rules deriving change verbs from adjectives or nouns via affixes such as *-ify* and *-ize*. The derived verbs, such as *humidify* and *magnetize*, generally refer to a state or attribute, and there is usually an opposite state or attribute value expressed by a verb that is identical except for a negative prefix (*dehumidify* and *demagnetize*). Many change verbs are also derived from adjectives via the *-en* suffix; they tend to have an antonymic verb derived from the base adjective's antonym: *weaken-strengthen, shorten-lengthen*, etc.

3. *Verbs of Communication*. The communication verbs, which comprise over 710 synsets, include verbs of verbal and nonverbal communication (gesturing); the former are further divided into verbs of speaking and verbs of writing. Most communication verbs involving language, such as *petition* and *hail*, can be classified as troponyms of speech act verbs (Austin, 1962). Verbs of verbal communication are richly lexicalized in English. They are elaborated in terms of manner of speaking (*lisp, stammer, babble*) or the speaker's intention (*beg, order, thank*). The communication verb file is also rich in denominals (*mandate, appeal, quiz*, and many more). Many sub-areas of lexicalization show where society values communication: politics (*veto, inaugurate*), law (*libel, plead, pardon*), religion (*preach, proselytize, catechize*), education (*teach, examine*), telecommunications (denominals derived from the nouns denoting the medium of communication, like *fax, telex*, and *e-mail*), to name just a few. This file also contains verbs referring to animal noises (*neigh, moo*, etc.) and verbs of noise production and uttering that have an inanimate source and lack a communicative function (*creak, screech*).

4. *Competition Verbs*. These verbs, grouped into over 200 synsets, cover the semantic areas of sports, games, and warfare. In this file, the *L-1* layer is relatively well lexicalized for many hierarchies, because there are many verbs referring to actions that are specific to games or sports, and that are troponyms of verbs with more general meanings. We find many composite troponyms (*face-off, run-off, counterstrike*) and denominals (*referee, handicap, arm, team, campaign, chicken-fight, duel*). Many of the verbs can be used reciprocally, i.e., they can take a plural subject, and either a surface or an implicit "each other" referring to the two opposing sides (*fight, race*, etc.).

5. *Consumption Verbs*. The consumption file includes, besides verbs of ingesting, verbs of using, exploiting, spending, and sharing, organized into approximately 130 synsets. Many verbs are syntactic unergatives—that is to say, they are either strictly intransitive or they select only cognate objects or their subordinates. For example, *drink*, when it is not used intransitively, takes as its object the cognate noun *drink* (usually with a modifier) or one of its subordinates (*water, beer*, etc.). The analysis of transitive and intransitive verbs like *eat* (Fellbaum & Kegl, 1989) described above shows how transitivity patterns can serve to tease out semantic subgroups of verbs. The consumption

verb file contains other unergatives like *eat*, which can be analyzed in a similar fashion.

6. *Contact Verbs*. This file is the largest verb file, consisting of over 820 synsets. Because most of its verbs are troponyms of very few base verbs, the largest tree structures are to be found in this file. The central verb concepts are {*fasten, attach*}, *cover, cut*, and *touch*. Many troponyms are derived via a manner relation encoding the force, intensity, or iteration of the action. For example, the troponyms of *rub* denote different degrees of force (*scrub, wipe, fray, chafe, scour, abrade*, and others). There are many verbs of holding (*grasp, squeeze, grab, pinch, grip*, and others), and touching (*paw, finger, stroke, hit, jab, poke*). Some of the base verbs require an instrument or material; Carter (1976) and Jackendoff (1983) refer to these as “entailed” or “open” arguments, respectively. These arguments are frequently conflated in troponyms together with the base verb characterizing the action. Troponyms of *cut* encode the instrument: (*knife, saw*) or the resultant shape (*cube, slice*); troponyms of *cover* express the material with which an object is covered (*paint, tar, feather*); troponyms of *enclose* encode the container (*box, bag, crate, shroud*). Verbs of removal refer to the removed stuff (*skin, bark, fleece*) or the resultant empty space (*furrow, hole, groove*). Body part denominals indicate what kind of contact action the body part is typically used for: *shoulder* (*support, carry*); *elbow* (*push*); *finger, thumb* (*touch, manipulate*), and so on.

7. *Cognition Verbs*. This file contains verbs denoting various cognitive actions and states, such as *reasoning, judging, learning, memorizing, understanding*, and *concluding*. The organizing relation in this file is troponymy. Some troponyms express kinds of reasoning (*deduce, induce*) or degrees of certainty (*infer, guess, assume, suppose*). The cognition verbs overlap to a large degree with the communication verbs; one verb can refer both to the mental activity of, say, reasoning and judging and to the action of articulating one’s reasoning and judging.

8. *Creation Verbs*. The creation verbs, organized into about 250 synsets, fall into several subgroups that are both semantically and syntactically motivated, but whose superordinates, referring to manners of creation, are not lexicalized: *create by mental act* (*invent, conceive*, etc.); *create by artistic means* (*engrave, illuminate, print*); *create from raw material* (*weave, sew, bake*). Many of these verbs can appear either transitively, where the direct object refers to the creation, or intransitively, where the verb no longer necessarily has the *create* sense but refers only to a manipulation of some material (compare *He sewed* and *He sewed a shirt*) without implying the accomplishment of a creation. Other troponyms of creation verbs are such denominals as *lithograph, fresco*, and *silkscreen*, where the created object has been conflated with the verb.

9. *Motion Verbs*. The motion verbs, grouped into over 500 synsets, derive from two roots: {*move, make a movement*}, and {*move, travel*}. The first sense, exemplified by *shake* and *twist*, is what Miller and Johnson-Laird (1976, p. 529) call “motion-in-place” (1976, p. 529) and Pinker calls “contained” motion (1989); the second is the concept of locomotion, as in *run* and *crawl*. Both senses of *move* can also have a transitive causative meaning (this is not true for all of their troponyms, though). Troponyms encode the speed of locomotion (*gallop, race*), the medium of transportation (*canoe, taxi*), and the medium in which the travel takes place (*fly, swim*), as well as other

elaborations.

10. *Emotion or Psych Verbs*. These verbs fall into two grammatically distinct classes: those whose subject is the animate Experiencer and whose object (if there is one) is the Source (*fear, miss, adore, love, despise*); and those whose object is the animate Experiencer and whose subject is the Source (*amuse, charm, encourage, anger*). In both cases, the Source may be either animate or inanimate. If the Source is animate, it may be either intentionally causing the emotion, i.e., it may an Agent, or it may be the unintentional Source of the emotion. This distinction is shown by the ambiguity of *The teacher frightened the children*. Inanimates are of course always unintentional sources: *The skeleton/The cry of the owl scared the children*. Most of the verbs have been structured along the analysis given in Johnson-Laird and Oatley (1989) for nouns, where five basic emotions (*happiness, sadness, fear, anger, disgust*) are posited along with their subordinates; most of these nouns have corresponding verbs. Besides being linked by troponymy, some emotions enter into opposition relations (*love-hate, hope-despair*).

11. *Stative Verbs*. Stative verbs (some 200 synsets) are for the most part verbs of *being* and *having*. Many stative verbs also have non-stative senses that have been placed into other files. For example, verbs like *surround, cross, and reach* have both a stative sense referring to spatial relations, and a non-stative sense denoting verbs of motion.

Many verbs have the meaning of *be* ADJECTIVE: *equal, suffice, necessitate*, to name just a few. Like adjectives, they usually have opposite terms (*differ, lack, obviate*) and synonyms (*match, cover, require*), but they rarely have superordinates, other than *be* or *{have, feature, have as a feature}*. The stative verb file, therefore, consists of small, semantically independent clusters, and resembles the adjective file.

12. *Perception Verbs*. The top nodes in this file of about 200 synsets are verbs referring to perception by means of the five senses. Their troponyms encode different elaborations. For example, the troponyms of the base verb *see* conflate the intention on the part of the perceiver (*watch, spy, survey*), the circumstances (*witness, discover*), and manner (*gaze, stare, ogle, glance*). Verbs of smelling, for example, take as their subject either experiencer (*snuff, sniff, whiff*) or the source of the perception (*reek, stink*); *smell* and *scent* can be used in either frame. Verbs denoting both the causation and the perception of cutaneous irritation are particularly richly lexicalized: *ache, hurt, prickle, sting, prick, tingle, tickle, scratch, itch, bite*, and others.

13. *Verbs of Possession*. The verbs of possession, organized into almost 300 synsets, are mostly derived from three basic concepts: *{have, hold, own}*, *{give, transfer}*, and *{take, receive}*, which denote the change of possession and its prior or resultant state. Troponyms of these verbs refer to ways in which the transfer of possession takes place in our society: by legal or illegal means, such as inheritance (*bequeath, will, inherit*) vs. theft (*rob, plagiarize, loot*); by formal or informal gifts (*bestow, confer, grant* vs. *beg, bribe, extort*); and by various business transactions (*peddle, scalp, auction, retail*), to name a few of the possible elaborations.

14. *Verbs of Social Interaction*. This file contains well over 400 synsets with verbs from different areas of social life: law, politics, economy, education, family, religion, etc. Many have a specialized meaning, restricted to a particular domain of social life, and

they tend to be monosemous (*impeach, court-martial, moonlight, franchise, gerrymander, excommunicate*, to name a few). Frequently, the verbs are denominals conflating a basic verb—often a communication or a cognition verb—and a noun from one of the areas named above, for example, *petition, quarrel, charm*, or *veto*.

15. *Weather Verbs*. Weather verbs constitute the smallest verb file (66 synsets), but they are semantically and syntactically distinct. They include mostly verbs like *rain* and *thunder*, which are all intransitives (except for such idiomatic expressions as *It is raining cats and dogs*). They do not select for any arguments; their subject is the semantically empty expletive, *it*. Many of these verbs are derived from their homonymous nouns (*rain, thunder, snow, hail*, etc.).

Conclusion

A relational analysis of English verbs has revealed some of the striking ways in which verbs differ from nouns and adjectives. The relations between verbs are distinct from those between words of other parts of speech; in general, their semantics are considerably more complex. The predominance of different relations and different lexicalization patterns in various semantic domains has been discussed.